

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Maximum [or fastest, highest] speed [or velocity] /constant speed [when forces are balanced] Accept: <u>speed</u> when forces are balanced	1			1		
		(ii)		Indicative content: Preparation 1. Set up the apparatus as shown, with the pointer (e.g.) 150 cm above the floor 2. Use a balance to determine the mass of a cake case [e.g. weigh 10 and divide by 10] Carrying out 3. Drop a single cake case from (e.g.) 20 cm above the pointer. 4. Using a stopwatch, record the time it takes to fall from the level of the pointer to the floor. 5. Repeat step 4 (another 4 times) and calculate the mean time. 6. Repeat steps 4 and 5 with a stack of e.g. 2, 3, 4 [and 5] cake cases. Analysis 7. Use the equation speed = distance/time to calculate the speed 5 – 6 marks Expect a full description of the method with at least 5 points mentioned from all areas. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		6

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				3 – 4 marks Expect an outline description of the method with at least 3 points made from two of the areas <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Expect 1 - 2 points made. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)		Substitution into $s = \frac{d}{t} = \frac{1.5}{0.6}$ (1) [or by implication] = 2.5 [m/s] (1)	1	1		2	2	2
		(ii)		4 points plotted correctly (ignore (0,0) to within <1 small square division (2) ecf on (b)(i) 3 points plotted correctly (ignore (0,0) to within <1 small square division (1) ecf on (b)(i) 2 or less points plotted correctly (ignore (0,0) to within <1 small square division (0) ecf on (b)(i) Smooth curve <u>through (0,0)</u> < 1 small square tolerance between curve and points (1)		3		3	3	3

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		(iii)		Pair of values of time (e.g. 1 and 2 cakes cases: 0.90 and 0.68 [s] or 2 and 4 cake cases: 0.68 and 0.56 [s]) (1) Calculation of ratio e.g. 0.8 / implied use of ratio e.g. half of 0.9 is 0.45 (1) Comparison to 0.5 and conclusion that suggestion is not supported (1) Alternative approach using speeds: Pair of values of speed (e.g. 1 case → 1.7 [m/s] and 2 cases → 2.2 [m/s]). (1) To halve the time means double speed (1) Comparison e.g. $2.2 \neq 2 \times 1.7$ / 2nd speed should be 3.4 and so suggestion not supported (1)			3	3	2	3
				Question 5 total	8	4	3	15	7	14